

QV

QUANTUMVAULT

Product & Architecture Master Plan

Competitive Landscape | Build Strategy | Team & Roadmap | GTM

A complete go-to-market strategy, engineering build plan, team composition model, and exhaustive competitive intelligence document for the QuantumVault Post-Quantum Token Infrastructure platform — prepared for Product Owner and Senior Technical Architect review.

This document covers: 30+ competitive products analyzed across 5 market segments, full engineering roadmap with team structures, cost modeling, risk register, technology stack decisions, and go-to-market sequencing.

Version 1.0 | April 2026 | CONFIDENTIAL — INTERNAL STRATEGIC

Prepared by: Product & Architecture Team

PART I — COMPETITIVE ANALYSIS

PART II — PRODUCT BUILD PLAN

PART III — GO-TO-MARKET STRATEGY

— PART I —

COMPETITIVE ANALYSIS

SECTION 1

1. Market Landscape Overview

The QuantumVault competitive landscape spans **five distinct market segments**. No single competitor operates across all five — and none has shipped a product with post-quantum cryptography, hardware binding, self-mutation, AND zero-knowledge in a unified token format. Our analysis covers 30+ products, their funding, their PQ roadmaps, and their weaknesses.

Segment	Key Players	Market Size (2026)	PQ Readiness	Our Overlap
Identity-as-a-Service (IDaaS)	Auth0/Okta, AWS Cognito, Azure AD	\$2.8B	NONE (roadmap only)	Token format replacement
Secrets & Vault Management	HashiCorp Vault, AWS Secrets Manager	\$890M	NONE production	Auth token complement
Post-Quantum Crypto Vendors	PQ Shield, ISARA, Sandbox AQ, PostQuantum Ltd	\$1.2B	PARTIAL (libraries only)	Direct category creator
Zero-Trust Network Access	Cloudflare Access, Zscaler, Google BeyondCorp	\$490M	NONE	Token layer replacement
Open Source / Standards	liboqs, SPHINCS+, pqcrypto, OpenSMAPQ	\$0	ALGORITHM ONLY	No token system exists

Key insight: The post-quantum crypto vendor segment is nascent (\$1.2B) and consists entirely of library and consulting vendors — no one has productized it into a complete, deployable token infrastructure. This is the white space QuantumVault occupies.

SECTION 2

2. Segment 1 — Identity-as-a-Service (IDaaS)

This is the largest adjacent market. These companies issue tokens (always JWT) to billions of users. They are not post-quantum, not hardware-bound, not self-mutating. Their PQ roadmaps are vague. They represent our future M&A; acquirers or our displacement targets.

Company	Funding / Valuation	Token Format	PQ Roadmap	Hardware Binding	Key Weakness	Threat to QV
Auth0 (Okta)	\$16.5B acq. (2021) Okta: \$10B market cap	JWT (RS256/HS256)	No public roadmap (CISA urged in 2025)	NO	JWT flaws inherited; cloud-dependent; vulnerable to HNDL	LOW — different layer; can be QV customer
AWS Cognito	AWS (\$2T company)	JWT (RS256)	AWS post-quantum TLS beta only	NO	Vendor lock-in; jwt all the way; no on-prem option	LOW — AWS customers need on-prem solution
Azure AD B2C	Microsoft (\$3T)	JWT + SAML	Azure PQ TLS; no token PQ	Conditional Access (policy, not crypto)	GDPR issues; no ZK proofs; JWT dependent	LOW — enterprise look for alternatives
Google Identity	Google (\$2T)	JWT (OpenID)	No token PQ; QKD research separate	NO	Google data dependency; no on-prem; no hardware binding	LOW
Ping Identity	\$2.8B (Thales acq.)	JWT/SAML	Exploring PQ (press release 2025)	NO (uses FIDO2 as separate layer)	Legacy architecture; expensive; no real PQ shipping	MEDIUM — exploring same space slowly
ForgeRock (Ping)	Merged into Ping	JWT/SAML/OAuth	None	None	Legacy; enterprise-only; complexity	LOW
Duo (Cisco)	\$2.35B acq. (2018)	JWT internally	Cisco PQ roadmap 2027 target	Device fingerprinting (not cryptographic)	MFA only; no ZK; no PQ tokens	LOW — MFA layer not token layer

Competitive conclusion (IDaaS): These are our best potential customers AND our best acquisition prospects. None will ship PQ token infrastructure before 2028 at earliest. Auth0/Okta, AWS, and Azure each have announced PQ TLS — but TLS is the transport layer; the token format (JWT) remains classically signed. QV solves the layer they are ignoring.

2.1 Detailed: Auth0/Okta — Most Relevant IDaaS Competitor

Dimension	Auth0/Okta Detail	QV Advantage
Token format	JWT (RS256 or HS256); SAML available	Binary PQ format; Dilithium-5; no algorithm negotiation
Post-quantum	No roadmap for token PQ; only TLS-layer PQ not	NIST FIPS 204/203 production-ready
Hardware binding	Device fingerprinting via browser heuristics (soft)	FIDO 2.0 + PUF cryptographic binding (hard)
Self-mutation	None — tokens valid until TTL	Every use mutates; replay architecturally impossible
Revocation	Redis blacklist (defeats stateless model)	Cryptographic accumulator — no DB needed
Zero-knowledge	None — all claims visible to all services	Bulletproofs per claim — GDPR-native
Dependencies	40+ npm packages; supply chain risk	Zero dependencies — Rust no-std core
Deployment	Cloud-only (okta.com domain); limited on-prem	Self-hosted; air-gapped supported
Price	\$23/MAU (enterprise)	TBD — see pricing model
Strength	Brand, ecosystem, 18,000+ integrations	We need to partner before we displace

SECTION 3

3. Segment 2 — Secrets & Vault Management

Vault tools manage secrets (API keys, DB passwords, certificates) but do NOT issue request-level authentication tokens. They are adjacent — not direct competitors — but customers consider them when evaluating QV. Key distinction: HashiCorp Vault stores secrets; QuantumVault authenticates requests.

Company	Product	What It Does	Crypto Used	PQ Status	Why Not a Token System
HashiCorp (IBM)	Vault	Secrets management, dynamic credentialing	AES-256 GCM, Ed25519	NONE — quantum vulnerable	No request-level auth tokens; requires Consul backend
AWS	Secrets Manager + ACM	Secret storage, certificate issuance	AWS KMS (classical)	AWS PQ TLS only	Cloud-only; no auth tokens; vendor lock-in
CyberArk	PAM + Conjur	Privileged access; vault; credentialing	AES-256 + RSA	NONE	Human PAM focus; very expensive (\$500K+); no managed
Azure	Key Vault	Key/secret storage, HSM-backed	RSA/ECDSA + AES	Azure HSM PQ in preview	Azure-only; no auth tokens; no ZK
Thales	CipherTrust	Enterprise key management, HSM	Classical crypto	FIPS 140-3 HSM work	Hardware HSM focused; very expensive; no PQ token

Strategy: Partner with HashiCorp/IBM and CyberArk rather than compete. QV solves the token authentication problem; they solve the secrets storage problem. A joint integration story is compelling to enterprise buyers.

SECTION 4

4. Segment 3 — Post-Quantum Crypto Vendors

This is our most direct competitive segment. These companies sell PQ cryptographic libraries, consulting, and crypto-agility tooling. None have shipped a complete token infrastructure product. They are potential partners (license their audited implementations) or acquirers (they want a productized token layer).

Company	Founded	Funding	Product	What They Ship	Token System?	Threat Level
PQShield (Oxford, UK)	2018	\$37M Series B (2023)	PQCrypto SDK, Hardware IP, SoC	Chip-level PQ crypto IP; SDK	NO — library only; for embedded; no token format	MEDIUM — could build token product; watch closely
ISARA (Canada)	2015	\$13M+	ISARA Radiate Toolkit	PQ migration toolkit; agility libraries	NO — migration libraries; consulting tooling only	LOW — consulting-led; not product company
Sandbox AQ (Google spin-out)	2022	\$500M+ (Google-backed)	AQtive Guard, AQtive Platform	PQ crypto platform; enterprise migration	NO — migration platform, not tokens	HIGH — Google money; could move to tokens fast
Post-Quantum Ltd (UK)	2009	Private	NTS-KEM, Hybrid communication	PQ communication protocol; VPN	NO — VPN/comm; not auth tokens	LOW
Crypto4A (Canada)	2018	Private	QASM HSM	PQ-ready HSM hardware	NO — hardware only	LOW — partner candidate
evolutionQ (Canada)	2016	Private	QKD migration	Quantum key distribution; network migration	NO — QKD focus	LOW — different layer
QuSecure (US)	2021	\$28M Series A (2022)	QuProtect Orchestration	PQ-TLS orchestration; crypto-agility at edge	NO — TLS only	LOW — TLS not tokens
IBM Quantum Safe	—	IBM (\$130B)	IBM Quantum Safe explorer	CBOM scanning; migration consulting	NO — consulting, no product	MEDIUM — IBM could build this overnight
Cloudflare Research	—	Cloudflare (\$16B)	PQ TLS beta, NIST submissions	PQ-TLS at CDN edge; no token support	NO — TLS only	HIGH — already at edge; if they add token PQ we compete

Critical insight — Sandbox AQ: Google-backed, \$500M+, expanding aggressively. They currently focus on enterprise PQ migration scanning, but a pivot to token infrastructure is a single product decision away. This is our most important competitor to watch. **We must ship a credible v1 before Sandbox AQ enters the token space.**

SECTION 5

5. Segment 4 — Zero-Trust Network Access (ZTNA)

ZTNA platforms enforce identity-based access at the network edge. They issue short-lived tokens but depend on JWT format and classical cryptography. They route all traffic through their infrastructure — a critical weakness for regulated industries.

Company	Product	Market Cap / Val	Token Format	PQ Status	Key Weakness
Cloudflare	Access + Gateway	\$19B	JWT (short-lived)	PQ-TLS at edge; no token PQ	All traffic via Cloudflare; no on-prem; JWT tokens
Zscaler	ZPA + ZIA	\$18B	JWT/SAML	Announced PQ TLS 2025; no token PQ	Cloud-only; expensive; no hardware binding
Google	BeyondCorp Enterprise	Google (\$2T)	JWT (GCP tokens)	PQ TLS experiments; no production token PQ	Google infrastructure dependency; US entities only
Palo Alto	Prisma Access	\$95B	JWT/SAML	PQ TLS roadmap 2026	Very expensive; complex; no ZK; no hardware to
Microsoft	Entra Private Access	Microsoft (\$3T)	JWT (AAD tokens)	Azure PQ preview	Azure-only; vendor lock-in; GDPR concerns
Tailscale	WireGuard mesh VPN	\$1B+	Auth keys (not JWT)	No PQ	Emerging; no enterprise compliance certs yet

Strategy vs ZTNA: These are not token competitors — they are distribution channels. QV tokens can be issued by their platforms as the underlying token format. A Cloudflare or Zscaler partnership that replaces JWT with QV tokens inside their platform is a billion-dollar distribution opportunity.

5.1 Emerging Competitor: Cloudflare — Watch List

Cloudflare is uniquely positioned and uniquely dangerous. They run the largest TLS termination network globally, have already deployed PQ-TLS in production, have a research team that contributed to NIST PQC submissions, and have deep relationships with every enterprise security buyer. If they decide to ship a PQ token format — they could do so in 6-12 months. Our moat against Cloudflare is: (1) on-prem/air-gapped deployment, (2) hardware binding (they have no TPM/PUF integration), (3) zero-knowledge proofs (not in their roadmap), and (4) DQA distributed authority (they ARE the centralized authority by definition).

SECTION 6

6. Segment 5 — Open Source & Standards Bodies

Project	Maintainer	What It Provides	Token System?	Our Relationship
liboqs	Open Quantum Safe (U.S. Army)	PQ algorithms implementations	NO — algorithms only	Validate our implementations against; potential
SPHINCS+	NIST / Academic consortium	Reference SPHINCS+ implementation	NO	We implement from FIPS 205 spec directly
pqcrypto (Rust)	Community	Rust bindings for liboqs	NO	We do NOT use (supply chain risk); implement
OpenSSL 3.x (OQS)	OpenSSL + OQS project	PQ cipher suites for TLS	NO — TLS only	Complementary; customers use both
PASETO	paragonie / community	Platform-agnostic token spec (v4)	YES — but classical	Only closest OSS competitor; we supersede it
Biscuit (Clever Cloud)	Community	Datalog-based auth tokens	YES — no PQ	Interesting attenuation model; our ZK is superior
Macaroons (Google)	Google Research (2014)	Contextual capability tokens	YES — no PQ; no encryption	Academic ancestor; no active development
IETF OAuth WG	IETF	JWT/OAuth standards; drafts for DPoP	NO/Not	Submit QV as IETF draft (strategy)
W3C WebAuthn	W3C	FIDO2 device-bound authentication	Authentication only (not authorization tokens)	Complementary; QV extends WebAuthn into tokens

PASETO is the only open-source token competitor. It is well-designed and respected. Strategy: engage the PASETO community early; position QV v1.0's Suite 0xF0 as the PQ upgrade path for PASETO v4 deployments. PASETO users are our warmest audience.

SECTION 7

7. Competitive Positioning Matrix

The following matrix positions all major competitors across the two axes that define our market: **Security Completeness** (x-axis: 1=JWT, 10=all 12 QV layers) and **Deployment Independence** (y-axis: 1=cloud-only, 10=fully self-hosted air-gapped).

Company / Product	Security Score (/ 10)	Independence (/ 10)	PQ Token Score (/ 10)	Combined	Verdict
QuantumVault v2.0	10	10	10	30/30	BENCHMARK
QuantumVault v1.0	9	10	9	28/30	STRONG
PASETO v4 (local)	5	9	2	16/30	Best current OSS — no PQ
Sandbox AQ	7	6	6	19/30	Watch — growing fast
PQShield SDK	7	8	5	20/30	Algorithms only; no tokens
HashiCorp Vault	6	8	1	15/30	Secrets mgmt; not tokens
Auth0 / Okta	4	3	1	8/30	Massive ecosystem; JWT-locked
AWS Cognito	3	1	1	5/30	Cloud-locked; JWT
Cloudflare Access	4	2	2	8/30	ZTNA; dangerous distributor
JWT (baseline)	1	7	0	8/30	Legacy; quantum-vulnerable
Biscuit / Macaroons	4	8	0	12/30	Interesting but no PQ

7.1 Threat Tier Classification

Tier	Companies	Rationale	Our Response
Tier 1 — Immediate Watch	Sandbox AQ, Cloudflare	Resources + PQ knowledge + distribution	Ship product v1 with enterprise customers before they do
Tier 2 — Strategic Partners	Auth0/Okta, CyberArk, HashiCorp	Not building tokens; need QV; will buy or build	Build integrations; pursue BD deals; position for a future acquisition
Tier 3 — Slow Movers	AWS, Azure, Google, Ping	Big co. inertia; PQ is 2028+ on their roadmap	Win enterprise customers they can't serve; build standards
Tier 4 — Non-Competitors	QuSecure, evolutionQ, PostQuantum	Different layers (TLS, QKD, VPN)	Partner for full-stack PQ story
Tier 5 — OSS Community	PASETO, Biscuit, liboqs	No commercial threat; warm audience	Engage community; open-source parts of QV spec

SECTION 8

8. Our Moat — Defensibility Analysis

A competitive moat is only as good as the time it takes to replicate. The following analysis estimates replication time for each QV capability:

QV Capability	Time to Replicate (well-funded team)	Why It's Hard	Moat Strength
Dilithium-5 + Kyber-1024 token integration	3-6 months	NIST specs are public; algorithms are standard	WEAK (threshold)
Self-mutating tokens (hash chain ratchet)	2-4 months	Novel but describable; someone will clone it	MEDIUM
Zero-knowledge Bulletproofs per claim	6-12 months	ZK engineering is rare; Bulletproof implementation is difficult	MEDIUM-STRONG
Cryptographic accumulator revocation (no DB)	4-8 months	Academic concept; implementation requires cryptomath expertise	MEDIUM
DQA Threshold signing (3-of-5 distributed)	8-14 months	Threshold Dilithium is non-trivial; DKG protocol complex	STRONG
Verifiable Delay Functions in token format	6-10 months	VDFs are new; calibration and verification subtle	STRONG
Physical Unclonable Function binding + enrollment	12-18 months	Requires hardware partnerships; fuzzy extractor expensive	VERY STRONG
Formal TLA+/Coq protocol verification	12-18 months	Rare skill set; certification process takes time	VERY STRONG
FIPS 140-3 certification of custom PQ module	18-30 months	NIST certification process is fixed-time; cannot be accelerated	VERY STRONG
Enterprise customer relationships + trust	Years	Security buyers are conservative; trust takes years	STRONGEST

Moat conclusion: The algorithmic capabilities can be replicated in 6-12 months by a well-funded team. Our durable moat is: (1) FIPS 140-3 certification (cannot be shortened), (2) PUF hardware partnerships, (3) formal verification credentials, and (4) the first enterprise customer relationships in the banking/defense sector. **Speed to market and enterprise trust are our primary moat — not the algorithms alone.**

— PART II —

PRODUCT BUILD PLAN

SECTION 9

9. Product Vision & North Star Metrics

"Be the authentication token standard for every organization that cannot afford to be wrong about security — banking, government, healthcare, defense."

9.1 Product Pillars

Pillar	Definition	Failure Condition
Unbreakable by Architecture	No patch needed — security is structural, not reactive	Any token forgery in production = pillar failure
Zero Operational Overhead	Drop-in JWT replacement; no new infrastructure required	Requires more than 2 config lines to integrate
Quantum-Ready from Day 1	Not "upgrade later" — PQ is the default, not the option	Shipping classical-only mode as primary
Formally Verified	Math-proven protocol, not just tested	No TLA+ model for core invariants at launch
Zero Supply Chain Risk	No dependencies to compromise	Any npm/pip transitive dependency in core

9.2 North Star Metrics (12-month targets)

Metric	Target (Month 12)	Target (Month 24)
Tokens verified per day (production)	100M	10B
Enterprise production customers	5	50
SDK languages supported	5 (Rust, Go, Python, JS, Java)	12+
Mean time to integration (new customer)	< 2 hours	< 30 minutes
Security CVEs in core	0	0 (forever)
FIPS 140-3 certification status	Submitted	Certified Level 2
Annual Recurring Revenue	\$2M ARR	\$20M ARR

10. Technology Stack Decisions

10.1 Core Cryptographic Engine — Rust (no_std)

Every decision in the core engine prioritizes correctness and auditability over developer convenience:

Component	Technology Choice	Rationale	Alternative Rejected
Language	Rust (stable, no_std compatible)	Memory safety by type system; no GC; WASM target; untrusted OS support.	C++: complexity.
Dilithium-5	Implement from NIST FIPS 202 spec	Zero supply chain risk; auditable; no crate dependency	any crate: supply chain risk
Kyber-1024	Implement from NIST FIPS 202 spec	Same rationale	oqs-rs: transitive deps
SPHINCS+-256	Implement from NIST FIPS 202 spec	Same rationale	sphincsplus crate: unaudited
XChaCha20-Poly1305	Pure Rust constant-time impl	256-bit nonce; no IV reuse risk; proven security	AES-GCM: IV reuse vulnerability
SHA3-256 / SHAKE-256	Pure Rust Keccak-f[1600]	Zero dep; FIPS 202; SIMD vectorizable	sha3 crate: acceptable but dep
VDF (Wesolowski)	Pure Rust big-integer arithmetic	No external math library needed; auditable	GMP bindings: C FFI risk
Serialization	Custom MessagePack impl	Binary-only; no JSON parser surface; zero dep	dep-serde: dep chain
SIMD acceleration	Rust std::arch (AVX-512 / NEON)	Compile-time feature flags; safe abstraction	Inline ASM: less portable
ZSTD compression	zstd-sys (C binding) OR pure Rust impl	Platform compression only; isolated from crypto path	any crate: heavy dep

10.2 SDK Layer Architecture

Target	Technology	Distribution	Timeline
Rust native library	Core engine (no_std)	crates.io (private registry initially)	Month 2
C FFI shared library	cbindgen-generated headers	.so / .dylib / .dll	Month 3
Go SDK	cgo bindings to C FFI	go get quantumvault.io/go	Month 5
Python SDK	cffi / ctypes over C FFI	PyPI private, then public	Month 5
Node.js SDK	napi-rs native addon	npm registry	Month 6
Java / Kotlin SDK	JNI over C FFI	Maven Central	Month 7
C# / .NET SDK	P/Invoke over C FFI	NuGet	Month 8
WebAssembly	wasm-pack + wasmbindgen	npm + CDN	Month 7
iOS / macOS	xcframework from Rust	Swift Package Manager	Month 9
Android	JNI from Rust (cargo-ndk)	Maven / AAR	Month 9
Embedded / IoT	Rust no_std binary	Direct integration	Month 10

10.3 Authority Server Infrastructure

Component	Technology	Why
API layer	Rust + Axum (async HTTP)	Same language as core; zero-cost async; type-safe
DQA consensus	Custom BFT protocol over mTLS	No existing BFT library fits; build lightweight purpose-built
QRNG integration	ID Quantique QRNG PCIe card + FHE	Hardware true randomness; IDQ is NIST-certified

Component	Technology	Why
HSM for key shards	Thales Luna HSM (FIPS 140-3 Level 3)	Key shards never in software memory
Kubernetes operator	Rust operator (kube-rs)	Cloud-native deployment; GitOps compatible
Admin dashboard	React + TypeScript (SPA)	Minimal surface; no SSR secrets
Metrics / observability	Prometheus + Grafana; OTEL tracing	Standard; no vendor lock
Database (accumulator state)	RedisDB (embedded) + SQLite fallback	No external DB dependency for single-node
Multi-node state sync	Custom CRDT-based replication	Eventually consistent accumulator state

SECTION 11

11. Engineering Team Structure

Founding principle: Hire for depth over breadth. Every engineer on the core team must be able to read the NIST FIPS specifications and implement correctly. We do not need 100 engineers — we need 15 exceptional ones.

Role	Count	Required Expertise	Hiring Difficulty	Est. Salary (USD)
CTO / Chief Architect	1	Cryptography PhD OR 10yr post-quantum systems, published research	VERY HARD	\$350-380K
Senior Cryptography Engineers	3	Lattice crypto implementation; NIST PQC; Extra level of verification	EXTREME	\$220-300K each
Rust Systems Engineers	4	no_std Rust; SIMD; FFI; WASM; embedded targets	HARD	\$180-240K each
Security Researcher / Red Team	1	Protocol attacks; formal threat modeling; C/C++ history	VERY HARD	\$200-260K
SDK Engineers (Polyglot)	3	Go, Python, Java, JS/TS; FFI integration; developer experience	MEDIUM	\$150-190K each
DevSecOps / Infrastructure	2	Kubernetes; HSM integration; air-gapped deployment	MEDIUM-HARD	\$160-200K each
Product Manager (Security)	1	Enterprise security products; banking/gov work; NIST familiarity	MEDIUM	\$180-210K
FIPS / Compliance Lead	1	FIPS 140-3 certification process; CAVP; cryptography module	VERY HARD	\$180-240K
TOTAL TEAM (Month 1-6)	16	—	—	\$3.2-4.1M / year (total comp)
Growth hire (Month 7-12)	+8	SDK, sales eng, solutions arch	MEDIUM	+\$1.4-1.8M / year

11.1 Where to Find Cryptography Engineers

Source	Type of Talent	Approach
NIST PQC competition participants	Drilium/Kyber/SPHINCS+ originals	Reach out directly; offer research credits; co-author papers
CRyS / IAIK / CWI research labs	Academic PQC researchers	PhD recruitment; research collaboration agreements
NSA / CISA (post-public sector)	Applied cryptography practitioners	Target 10-15yr veterans; security clearance a bonus
Open Quantum Safe (liboqs contributors)	PQ implementers; practical focus	Already writing the algorithms; convert to product engineers
Black Hat / DEF CON / CRYPTOCONF	Security researchers; red teamers	Sponsor talks; recruit from the audience
Cloudflare / Google Research	Industry PQ practitioners	PQ TLS team members; understand scale
Jane Street / Two Sigma (Rust type)	Rust type formal methods experts	Competitive on salary; mission-driven pitch

12. Phase-by-Phase Build Roadmap — 18 Months

12.1 Phase 1 — Foundation (Months 1–4)

Goal: Working binary token parser + Dilithium-5 sign/verify + test suite. No authority server yet. Just the token format specification and its Rust implementation.

Week	Deliverable	Owner	Definition of Done
1–2	Binary token format spec v0.9 (RFC-style)	CTO + Product	All field offsets, encoding rules, version semantics documented
3–4	Rust core crate scaffold (no_std, zero deps)	Lead Rust Eng	cargo build --target thumbv7m-none-eabi succeeds; zero deps
5–8	Dilithium-5 from NIST FIPS 204 spec	Crypto Eng 1+2	KAT vectors pass; constant-time verified (duddect)
5–8	Kyber-1024 from NIST FIPS 203 spec	Crypto Eng 2+3	KAT vectors pass; constant-time verified
9–10	XChaCha20-Poly1305 + SHA3-256 impl	Crypto Eng 1	NIST test vectors pass; SIMD build flag works
9–12	Token serialize / parse / sign / verify	Rust Eng 1+2	Round-trip test: 100K tokens; zero panics; no alloc path
11–12	MessagePack serializer (no dep)	Rust Eng 3	Spec-compliant; fuzz-tested 10M random inputs
13–14	Mutation engine (hash chain ratchet)	Rust Eng 1	Replay test: counter N presented twice = auto-revoke
15–16	C FFI bindings (cbindgen)	Rust Eng 4	C test program issues + verifies token; valgrind clean
15–16	TLA+ spec: mutation counter invariant	Security Researcher	TLC model checker: no counter violations in 10^6 states

12.2 Phase 2 — Core Security (Months 5–8)

Goal: All 12 security layers implemented. Accumulator revocation. ZK proofs. Hardware TPM binding. VDF. PUF enrollment protocol. Alpha token format frozen.

Month	Deliverable	Owner	Definition of Done
5	Bulletproofs ZK per claim	Crypto Eng 1+3	Prove age>18 without revealing age; NIZK soundness test
5	RSA/Bilinear Accumulator revocation	Crypto Eng 2	Revoke token; witness invalid; no DB query needed
6	TPM 2.0 device binding (Linux + Windows)	Rust Eng 2	TPM2_Certify integration; attestation chain verifies
6	Apple Secure Enclave binding (macOS/iOS)	Rust Eng 3	Secure Enclave keygen; DeviceCheck attestation
6	Android StrongBox binding	Rust Eng 4	Android Keystore; Hardware attestation verified
7	Wesolowski VDF implementation	Crypto Eng 1	Sequential property test: parallel compute gives no speedup
7	PUF enrollment protocol + fuzzy extractor	Security Researcher	Enrollment + recovery with <0.1% error rate; ARM TrustZone
7	SPHINCS+-256 from NIST FIPS 205	Crypto Eng 3	KAT vectors pass; triple-sign mode (Suite 0xFF) works
8	DQA threshold signing (3-of-5 Dilithium)	Crypto Eng 2+3	Shamir DKG; 3-of-5 combine; single key never assembles
8	PCS HKDF ratchet epoch	Rust Eng 1	Forward + backward secrecy: compromised K_n != derive K_{n+1}
8	AVX-512 + NEON SIMD acceleration	Rust Eng 2	3x speedup on Ice Lake vs scalar baseline
8	Alpha security audit (external firm)	CTO	NCC Group or Trail of Bits; critical findings: 0

12.3 Phase 3 — SDK Ecosystem (Months 9–14)

Goal: Language SDKs, developer portal, first pilot customer, authority server production-ready.

Month	Deliverable	Owner	Note
9	Go SDK + Express.js middleware	SDK Eng 1	JWT drop-in replacement; < 10 line integration
9	Python SDK + FastAPI / Django middleware	SDK Eng 2	pip install quantumvault; decorator-based
10	Node.js (napi-rs) + WASM SDK	SDK Eng 3	npm install @quantumvault/node; zero transitive deps
10	Java / Spring Boot SDK	SDK Eng 1	Maven artifact; Spring Security integration
10	Authority Server v1 (single node)	Rust Eng 3+4	Kubernetes operator; OIDC bridge; admin dashboard
11	Authority Server DQA mode (3-node)	Crypto Eng 2	Docker compose for 3-node local test
11	JWT migration bridge (read JWT, issue QV)	SDK Eng 2	Zero-downtime migration path for JWT users
12	First pilot customer (banking/fintech)	Product + CTO	One enterprise processing 1M+ tokens/day in staging
13	Developer portal + documentation site	Product + SDK Eng	Interactive playground; spec reference; tutorial
13	C# / .NET SDK	SDK Eng 3	NuGet; ASP.NET Core middleware
14	Batch verification GA	Rust Eng 1	10K batch verify in < 100ms; benchmark published
14	SOC 2 Type II audit begins	Compliance Lead	Engage auditor; evidence collection starts

12.4 Phase 4 — Enterprise & Certification (Months 15–18)

Goal: 5 enterprise customers live, FIPS 140-3 submitted, Series A fundraise, formal verification complete.

Month	Deliverable	Status
15	FIPS 140-3 Level 2 submission to CMVP	Cryptographic Module Validation Program submission
15	ProVerif authentication model complete	Formal proof: QV token authentication is sound
16	ISO 27001 certification	ISMS certified; required for EU banking customers
16	HSM integration (Thales Luna / Entrust)	Key shards in FIPS 140-3 Level 3 hardware
17	iOS + Android SDK GA	Mobile token lifecycle; PUF via Secure Enclave
17	5 enterprise customers (>1M tokens/day each)	Minimum \$500K ACV each
18	Series A fundraise close	\$20–30M target; tier-1 VC; strategic investor (defense/banking)
18	Common Criteria EAL4+ evaluation begins	Government / military use case unlocked

SECTION 13

13. Cost Model & Funding Requirements

13.1 18-Month Budget Summary

Category	Month 1–6	Month 7–12	Month 13–18	Total
Engineering salaries (16 → 24 FTE)	\$2.1M	\$2.8M	\$3.4M	\$8.3M
Hardware (HSM, QRNG, dev kits)	\$320K	\$80K	\$120K	\$520K
Cloud / infrastructure	\$60K	\$120K	\$240K	\$420K
Security audits (NCC Group / Trail of Bits)	\$300K	\$200K	\$400K	\$1.0M
FIPS 140-3 certification process	\$0	\$150K	\$300K	\$450K
Legal / IP / patents	\$120K	\$80K	\$80K	\$280K
Recruitment / HR	\$200K	\$150K	\$100K	\$450K
Marketing / DevRel / conferences	\$50K	\$200K	\$400K	\$650K
Travel / enterprise sales	\$30K	\$150K	\$300K	\$480K
Contingency (15%)	\$490K	\$593K	\$755K	\$1.84M
TOTAL	\$3.77M	\$4.52M	\$6.10M	\$14.4M

13.2 Funding Strategy

Round	Amount	Timing	Use of Funds	Target Investors
Pre-seed / Founder	\$50K–1M	Now (Month 0)	First 3 engineers, MVP spec, hardware	Founders + angels with security expertise
Seed Round	\$5–8M	Month 3–4	Full founding team (16 FTE); Phase 1+2 complete	Sequoia, Crypto, a16z crypto/security, Bessemer
Series A	\$20–30M	Month 16–18	Scale GTM, SDK ecosystem, FIPS cert, 24 FTEs	Same VCs + strategic: Palantir, In-Q-Tel, Lockheed M

Strategic investor rationale: In-Q-Tel (CIA venture arm) and similar defense-adjacent investors provide both capital and credibility for government/defense sales — a market where trust is currency. Their involvement signals to DoD/NSA/GCHQ procurement officers that QV has been vetted. This is worth more than the capital itself.

SECTION 14

14. Risk Register & Mitigations

Risk	Probability	Impact	Mitigation
Sandbox AQ ships a PQ token prod more us	MEDIUM	HIGH	Ship pilot customer within 12 months; open-source the spec (not the impl) to cre
Dilithium/Kyber broken by new ma	VERY LOW	CRITICAL	Suite 0x0A / 0xFF fallback; SPHINCS+ survives any lattice break; crypto-agility a
FIPS 140-3 certification takes longer than expected	LOW	MEDIUM	Sell to non-FIPS-required customers first (EU, startups); FIPS is requirement for
PUF hardware partnerships fail	MEDIUM	MEDIUM	PUF is Layer 10 — optional enhancement; TPM binding (Layer 4) is sufficient fo
Not enough cryptography engineers available	LOW	HIGH	Remote-first globally; approach NIST PQC submission authors directly; partner v
Token size (2.6KB) rejected by customers	MEDIUM	MEDIUM	ZSTD compression reduces to ~2.0KB; batch mode amortizes; educate: 2.6KB a
Auth0/Okta builds PQ tokens internally	LOW	MEDIUM	Their JWT codebase is too intertwined; would take 2+ years and break 18,000 in
Regulatory requirements change (PQC delayed)	LOW	LOW	QV is already valuable without regulation; HNDL threat is real regardless of man
Customer demands source code escrow	LOW	LOW	Acceptable; spec is language-agnostic; escrow the Rust core; customer can imp
Core implementation has a logic vulnerability	MEDIUM	CRITICAL	TLA+ formal verification; external audit (NCC/Trail of Bits); bug bounty program;

— PART III —

GO-TO-MARKET STRATEGY

SECTION 15

15. Target Customer Profiles (ICP)

Profile	Industry	Size	Why They Buy QV	Entry Point	ACV Target
Tier 1A (Primary)	Banking / Capital Markets	>\$10B assets Top 100 globally	HNDL threat; PCI DSS 4.0; regulatory pressure	CISO, Chief SWFT Officer	\$200K-\$1M
Tier 1B (Primary)	Government / Defense Contractors	Federal agency or Top 20 primes	NSA CNSA 2.0 mandate; classified info	CTO, Chief Data Officer	\$100K-\$2M
Tier 2A (Secondary)	Healthcare / Pharma	\$1B+ revenue; HIPAA-regulated	Patient data HNDL risk; HIPAA + HITECH	Chief Info Officer	\$50K-\$100K
Tier 2B (Secondary)	Critical Infrastructure (Energy, Utilities)	National grid or top-tier utility	NERC CIP compliance; OT/IT convergence	IT Director	\$150K-\$500K
Tier 3 (Long-term)	Fintech / Crypto Exchanges	Series B+ startup; >\$100M valuation	Developer-centric; security differentiator	Bottom management	\$20K-\$100K

15.1 Anti-ICP — Who Not to Sell To

Profile	Reason to Avoid
Early-stage startups (<Series A)	Won't pay; integration bandwidth too low; reference value low
Mid-market SaaS (no compliance requirements)	Don't feel the pain; JWT is "good enough" for them
Consumer apps (social, e-commerce)	Token size/latency matters; not security-first buyers
Pure cloud companies (AWS/Azure native)	Wait for AWS/Azure to ship PQ; not independent

SECTION 16

16. Pricing Architecture

Tier	Product	Price	Includes	Target Customer
Open Core	QV Spec + Rust reference impl	FREE (Apache 2.0)	Binary token format; Suite 0x05 only; community support	Devtoppers, fintechs; OSS projects
Developer	QV Node + single authority	\$999/month (\$10K/year)	Up to 10M tokens/month; 5 SDKs; email support; Suite 0x05/0x06	Startups, SaaS companies
Professional	QV Cluster (3-node HA)	\$5K/month (\$50K/year)	Unlimited tokens; all SDKs; Suite 0x09/0xFF	SMB, mid-market, phone support
Enterprise	QV Enterprise + DQA	\$25K–100K/month	DQA 5-node; Suite 0xFF; HSM integration; FIPS 140-3 compliant; 24/7 support	BBK, large corp, 24/7 support
OEM / Platform	Embed QV in your product	Custom (% of rev)	White-label; SDKs included; joint GTM; co-selling	Okta, Cloudflare, ZTNA vendors
Government	QV Federal Edition	\$500K–\$2M/year	FISMA-ready; IL4/IL5; FIPS 140-3 Level 3; CSF v2.0	Govt, DoD, Fed agencies, personnel support

Open Core Strategy: Open-sourcing the specification and the Suite 0x05 reference implementation serves as our developer acquisition funnel. Developers who integrate QV for free become internal champions who push for the Enterprise tier. The spec being open also creates community pressure on Auth0/Okta to support QV format — positioning us as a standard-setter rather than a niche vendor.

16.1 Revenue Model Projections

Scenario	Month 12	Month 18	Month 24	Month 36
Conservative	\$500K ARR (5 customers)	\$2M ARR (15 customers)	\$5M ARR (35 customers)	\$15M ARR (80 customers)
Base Case	\$2M ARR (10 customers)	\$8M ARR (40 customers)	\$20M ARR (80 customers)	\$60M ARR (200 customers)
Aggressive (QV becomes standard)	\$5M ARR (25 customers)	\$20M ARR (100 customers)	\$60M ARR (250 customers)	\$200M ARR (700 customers)

SECTION 17

17. Sales Motion & Partnership Strategy

17.1 Sales Motion by Tier

Tier	Motion	Sales Cycle	Key Steps
Open Core → Developer	Developer-Led Growth	1–14 days	GitHub star → docs → npm install → upgrade prompt
Developer → Professional	Sales Assist	2–8 weeks	Usage alert → outbound SDR → demo → POC → close
Professional → Enterprise	Enterprise Sales	3–9 months	CISO meeting → security eval → compliance review → legal → close
Government	Channel (In-Q-Tel / SAIC / Bionics)	6–18 months	Partner-led; QV provides technical depth; partner owns relationship
OEM / Platform	Business Development	3–12 months	CTO/VP Product → joint pilot → revenue share agreement → GTM

17.2 Strategic Partnerships (Priority Order)

Partner	Type	Value Exchange	Priority
Thales / CipherTrust	Technology	QV uses Thales HSM; Thales bundles QV with HSM	P1 — Month 6
Cloudflare	Distribution	QV tokens as Cloudflare Access token format option	P1 — Month 10
HashiCorp (IBM)	Integration	QV issues auth tokens; HashiCorp stores their secrets	P1 — Month 8
ID Quantique	Technology	QV uses IDQ QRNG hardware; IDQ bundles QV	P2 — Month 4
Ping Identity	OEM / Acquisition	Ping acquires or white-labels QV for their enterprise customers	P2 — Month 12+
SWIFT	Ecosystem	QV as recommended token format for SWIFT member institutions	P3 — Month 18+
NIST / IETF	Standards	Submit QV binary format as IETF draft standard	P3 — Month 14+
MITRE / CISA	Government	QV on CISA recommended products list for PQC migration	P3 — Month 18+

SECTION 18

18. Launch Sequencing

Launch	Timing	What Ships	Success Metric
Alpha (Private)	Month 5	Binary spec + Rust library; Suite 0x05 only; 3 design partners	3 design partners integrating; weekly feedback loop
Beta (Invite-only)	Month 10	Full 12-layer implementation; 5 SDKs; authority & security	10 companies in beta; 1M tokens/day processed
v1.0 GA	Month 12	Production-ready; Node/Go/Python/Java/WASM; 1M keys; Developer Hub	1M keys; Developer Hub; 1K stars; first paying customers
v1.0 Enterprise	Month 15	DQA mode; HSM integration; FIPS submission	FIPS 140-2 Type 1 enterprise deal signed
Standards Submission	Month 15	IETF draft for QV binary format; GitHub spec	IETF draft accepted for working group consideration
v2.0 GA (PUF + VDF)	Month 18	All v2.0 capabilities; Suite 0xFF; formal verification	Government defense pilot; Series A close

18.1 Developer Community Strategy

Community is our free distribution channel. The following tactics build it systematically:

Tactic	Platform	Target	Timeline
Open-source the spec (Apache 2.0)	GitHub	500 stars in 90 days; community implementation	Month 1-3
QuantumVault Explained blog series	Dev.to, Hacker News, Substack	50k reads; NIST PQC accessible primer	Month 8-12
Conference talks: why JWKE is bad, Black Hat, QCon	DEF CON, Black Hat, QCon	Positive category thought leaders	Month 6-14
Interactive token inspector (Web)	Quantumvault.io playground	Developers build their first QV token in 5 min	Month 11
Bug bounty program	HackerOne	Security community validates our implementation	Month 12 (GA)
University research grants MIT, Stanford, ETH Zurich	Academic	Academic papers citing QV as reference	Month 18

19. Success Metrics & OKRs

19.1 Q1–Q2 OKRs (Months 1–6) — Build Foundation

Objective	Key Results	Measurement
Ship a credible, auditable crypto core	KR1: All NIST KAT vectors pass for Dilithium-5, Kyber-1024, SPHINCS+ KR2: Zero critical findings in external audit KR3: Constant-time verified (dueduct) for all signing paths	Automated CI test suite; audit report
Establish technical credibility	KR1: CTO presents at one tier-1 security conference KR2: 2 design partners signed (Fortune 500 security firms) KR3: TLA+ spec published and reviewed by 3 external cryptographers	Conference acceptance; signed NDAs; expert review letters
Build the core team	KR1: 3 cryptography engineers hired with PQ publications KR2: 2 senior Rust engineers hired with no_std experience KR3: FIPS compliance lead hired	Offer letters signed

19.2 Q3–Q4 OKRs (Months 7–12) — Prove Market

Objective	Key Results	Measurement
Ship production-ready product	KR1: 5 SDKs available (Go, Python, Node, Java, WASM) KR2: Authority server passes 1M tokens/day load test KR3: Mean integration time < 2 hours (measured with beta users)	Load test reports; user interviews
Validate ICP and pricing	KR1: 3 paying customers (>\$50K/year each) KR2: NPS from beta users > 50 KR3: First banking customer in production	CRM records; NPS survey; production traffic
Build market presence	KR1: GitHub: 1,000 stars KR2: 2 case studies published (with customer permission) KR3: Security press coverage in 3 tier-1 outlets (Wired, Ars Technica, Dark Reading)	GitHub analytics; case study PDFs; media coverage

19.3 Year 2 OKRs (Months 13–24) — Scale

Objective	Key Results
Enterprise market leadership	5 enterprise customers (>\$200K ACV); 1 government pilot; FIPS 140-3 certified
Ecosystem growth	12 SDK languages; 3 framework integrations; 1 major partnership (Cloudflare or HashiCorp)
Financial milestone	\$8M ARR; Series A closed (\$20–30M); 24-month runway
Standards influence	IETF draft submitted; cited by NIST PQC migration guidance; 1 academic paper citing QV

"The window is open. NIST finalized PQC standards in August 2024. No enterprise-grade PQ token infrastructure exists. Nation-states are harvesting encrypted traffic today. The organizations we are building for will need this — the question is only whether they get it from us, or wait in panic for their existing vendor to patch a decade of cryptographic debt. We build it. We ship it. We define the standard."

Attribute	Value
Product	QuantumVault — Post-Quantum Token Infrastructure

Attribute	Value
Category	New category: Post-Quantum Token Infrastructure (PQTI)
Primary ICP	Banking (Top 100 globally), Government (US/EU), Defense contractors
Build Timeline	18 months to Enterprise GA; 12 months to v1.0
Team (Month 6)	16 FTE; 4 cryptography engineers; Rust core; FIPS compliance
Funding Required	\$14.4M over 18 months; Seed (\$6M) + Series A (\$20M)
Moat	FIPS 140-3 certification + PUF hardware partnerships + first enterprise trust
Highest Threat	Sandbox AQ (Google-backed, \$500M+, expanding aggressively)
Key Partnership	Cloudflare (distribution) + Thales (HSM) + HashiCorp (complementary)
North Star Metric	10B tokens/day verified; \$20M ARR by Month 24
Open Source Strategy	Spec + Suite 0x05 reference impl open (Apache 2.0); Enterprise features commercial
Classification	CONFIDENTIAL — Internal Strategic April 2026